

Blinded-V5: the LGI-based compute module

design rationale, construction, parameters, and measurements

local-mixing project

2026-09-04

Where it fits in the pipeline

A secret circuit C on n wires is protected in two steps: it is first **gadgetized** into an equivalent but structurally-scrambled circuit on $2n$ wires, which is then **mixed** (fmix). Blinded-V5 lives inside the gadgetizer, and this document assumes you know the earlier gadgetize designs but nothing about V5.

The **gadgetize module** is a **5-stage** pipeline, each stage on $2n$ wires (data $0..n$, band/junk $n..2n$):

1. **slice** — a junk-guard zero-slice keyed on the band (dead at the input port, where the band is 0);
2. **seed the band wires** — each band wire set to $x_i \wedge \neg x_j$ from the honest inputs;
3. **compute** — realise the input circuit A on the $2n$ wires so the data half carries A 's output and the band is only read;
4. **re-seed / re-randomise the band wires** — band turnover;
5. **re-slice** — the closing junk-guard slice (fires at the output port).

One term to keep straight: the circuit A that the gadgetizer computes is **not** C **itself**. It is the **sliced sandwich of C** — the output of a *separate sandwich module* that wraps C (with a random D , interleaved slice blocks, a floated column) into a $2n$ -wire circuit with $A(x, 0) = (\text{junk}, C(x))$. The sandwich *generates* the circuit that the gadgetize module then computes and mixes; it is a different module and is not one of the five stages above.

Blinded-V5 is a drop-in replacement for stage 3 — the compute. Where the earlier compute stages (the drip **route_fire**, the product-share carriers) realise A by *routing* each operand into position and firing a gadget there, Blinded-V5 instead embeds A **inside one enormous masking identity**: the compute's body is a cloud of long masking identities on the band, and A 's gates fire from *within* the mask by a momentary, reversible unmasking of their operands. It takes A and emits an equivalent A' on the $2n$ wires; the other four stages are unchanged. (V5 re-randomises the band *internally* as part of its own masking hygiene — §2 — which is distinct from the separate stage-4 re-seed.)

Source: `src/preprocessing/blinded.v5.rs`. Driver: `gen_sandwich_gadget ... blinded-v5;`
pipeline: `gss_mix.sh --gadgetization-mode blinded-v5`. The Markdown twin of this document is `BLINDED_V5_LGI_DESIGN.md`.

1 Why this design — the rationale

The compute stage must move the intermediate state as far as possible from the plaintext computation A while keeping A' exactly equivalent to A on the data wires. The design is a chain of forced moves.

(a) **Long identities are the only thing that moves the state.** Across the HD and affine heatmaps, short local edits barely displace the state; the ridge of recoverable structure only recedes when the masking identity woven through a wire is *long* — comparable to the circuit itself.

(b) **The only long identity that moves the state is the commutative structure of an identity with a single active wire.** A run of gates all targeting one active wire w and reading only controls is an identity iff the increments XOR to zero, and *all its gates commute* — which lets it be arbitrarily long and freely interleaved.

(c) **Entangle identities on different active wires by sharing the control wires — the band.** Every identity targets a data wire and reads only the band ($n..2n$), so identities on different active wires also commute; a band wire participates in the masks of many data wires at once.

(d) **Add random updates of the control wires from the active wires — in bursts.** Band updates **re-randomise** the controls (fresh band entropy throughout the run, not just at the seed) and **break naive commutation-back** (once a band wire is updated mid-run, the identities reading it before and after no longer commute past it), without ruining correctness (each straddling identity is closed before the update or re-derived across it). The updates are emitted in **bursts**: one *slot* is $F \approx 8K$ gates all targeting a single band wire b , each control drawn independently as a live-data or a band wire (so a burst spans $b \oplus= data \wedge aux$, $b \oplus= aux \wedge aux$, and $b \oplus= data \wedge data$). Bursting **concentrates** the band mixing into few slots (so fewer straddle-closures interfere with the LGIS) and gives each band wire a **data-wire-like burst-of-activity signature**.

(e) **Embed A inside one circuit-spanning identity via hidden unmasking.** Each gate fires from inside the mask: its operands are momentarily and reversibly unmasked into a linear combination of band wires (never bare), it fires as an expansion over those masked wires, then re-masks.

That is the whole design: a long, entangled, self-updating masking cloud with A firing from inside it. Everything in §2 is *how* to realise (a)–(e) as a single correct circuit.

2 The construction

A has n wires and m gates; A' has $2n$. Let u_w be the uses of wire w in A and w_w its writes.

Masking atom. $g57(w, x, y) = w \oplus= 1 \oplus (\neg x \wedge y)$ (data target, band controls), i.e. the mask term $1 \oplus y \oplus xy$. A disjoint-pair LGI $\bigoplus_i g57(w, cy_{2i}, cy_{2i+1})$ is a deg-2 mask; a **g57** and its reverse **linearise**, $g57(w, r_1, r_2) \oplus g57(w, r_2, r_1) = w \oplus r_1 \oplus r_2$ — the original read exploited this; the current read (**quad-fire**, step 2 below, default since 2026-09-06) deliberately does *not*, because a linearised operand is exactly affine in band wires for a window that the pipeline’s reordering stages stretch into a leak (§5.0). Disjoint pairs, not a K -cycle (which would telescope to a bare operand); $K \geq 2$.

Why one pass — co-sampling. The masks, the band updates, and A ’s gates are **co-sampled**: produced together in a single forward pass, rather than laying a fixed masking scaffold first and threading A through it afterwards. The reason is hidden firing (§3): every A -gate must be straddled by a *real*, rerand-protected masking identity opened at the exact instant the gate fires. Only a co-sampled pass can open that identity on demand, so the firing-hiding is complete while reusing the wire’s existing mask budget — at no size cost. (A fixed scaffold, filled in afterwards, starves at the tail and covers only 60–70% of gates; §3.)

The algorithm. A cheap setup, then a forward pass.

Setup.

1. **Mask budget.** Give each active wire w exactly $u_w + 1$ masking identities, split into w_w **straddle opens** (one per write to w , opened *on demand* at the gate that writes it — for hidden firing) and $\text{reads}_w + 1$ **filler opens** (opened between gates to keep w masked while it is read). At most **max_open** are open on a wire at once. This is the same budget a fixed scaffold would use, so the statistics are unchanged.
2. **Order.** Build the data-hazard DAG of A (**compute_deps**): a read is ordered after every earlier write of the wire, and a write after every earlier read — but same-target XOR writes commute, so there is *no* write-after-write edge. Any linear extension preserves A ’s function; a Kahn **ready queue** holds the gates whose predecessors are all placed.
3. **Rerand plan.** A shuffled list of **straddle_slots** + **repair_slots** band-update *slots* (§1d; auto **straddle_slots** $\approx m/(4K)$, **repair_slots** = 0).
4. **Rate.** Fire one rerand slot every $\text{total_steps}/\#\text{slots}$ primitive steps (a *step* = one A -gate placement or one filler open; $\text{total_steps} = m + \Sigma \text{filler}$), so the slots spread evenly and the last lands *during* the pass — no end-flush (a surplus of requested slots over total_steps warns rather than truncating silently).

Forward pass — repeat until all m gates are placed:

1. **Pop** a ready A -gate g (target wire c).
2. **Masked read (quad-fire).** For each operand wire, leave its currently-open masks *as they are* and describe the wire as the ANF polynomial $P_w = w' \oplus \sum_{(x,y)} (1 \oplus y \oplus xy)$ over its net-open pairs (a pair whose reverse is also open contributes the linear $x \oplus y$); top up with **fresh single g57s** (quadratic, not pair-completed) until at least **max_open** quadratic terms mask the operand (the **hard floor**: never bare, never affine). Remember the top-up gates as the *undo* — except that an operand with **no open lgi at all** keeps its first top-up as a real LGI (the *read-cover* rule, §5.6: otherwise a seldom-used wire returns to holding its plain value for the whole idle stretch after the fire). (Legacy linear read, **BV5_QUAD_FIRE=0**: complete each open **g57** with its reverse so the wire carries $\text{operand} \oplus \rho$, ρ a linear XOR of band wires, top up with pairs until $|\rho| \geq \text{min_mask}$, undo the reverses after the fire — see §5.0 for why it was replaced.)
3. **Fire from inside the mask.** Expand $c \oplus = \text{comp} \oplus \text{lit}(a) \wedge \text{lit}(b)$ as the polynomial product $P_a \cdot P_b$ (a negative literal complements the constant) into a batch of monomials over **ONLY** the masked control wires and band wires — never bare a , b , or $a \oplus b$ (0/1/2 controls and all polarities; the 0-control fire is $\neg \text{comp}$). At $K = 2$, **max_open** = 3 that is up to $8 \times 8 = 64$

monomials of degree ≤ 4 (versus 49 of degree 2 plus the linearise/undo bracket in the legacy read), so the gadget is $\sim 12\%$ *smaller*.

4. **Straddle (hidden firing, §3).** Split the monomial batch into two halves, shuffle each **independently** (they all target c and commute), emit the first half, **open a fresh lgi on c here** (one of its straddle opens, generated on demand), then emit the second half. The mid-fire open makes the module's net XOR on c equal $\Delta \oplus \text{secret-mask}$, not the bare increment Δ . The whole block is **bracketed by a temporary cover mask on c** drawn away from every band wire of the operands' polynomials (opened before the first monomial, closed after the last; the straddle open is drawn away from those wires too): a band wire shared between one of c 's masks and a fire monomial would cancel or fold that mask's uniform term and leave the mid-fire segments biased toward c_{new} (§5.7; ~ 6 gates per fire, $\approx 3\%$).
5. **Undo** — emit the top-up gates again (each is an involution), restoring the operands' original mask set.
6. **Maybe rerand** — if the calibrated rate says so, emit one band-update slot (below).
7. **Release & fill** — decrement the in-degree of g 's dependents, moving any that reach zero into the ready queue; then open a few **filler** LGIs on wires chosen by remaining filler budget.

Finish. Drain any remaining filler opens, then close every still-open LGI so the data wires end holding A 's true output.

The rerand slot (step 6) is a §1d burst: one band wire b , $F \approx 8\text{K}$ updates. A STRADDLE slot first **closes** the masks reading b (this is what thins masking past the ≈ 1024 knee — hence keeping **straddle_slots** at that budget); a REPAIR slot brackets the burst with each such mask's b -reading g57 (old b cancels, new b re-adds, so the mask stays open — no thinning). Running in the same pass, the band updates protect the straddle opens automatically. **Coverage across the burst is a rule, not a statistic:** when the masks reading b are *all* of a wire's open masks, the slot first opens a *replacement* LGI on that wire (sampled away from b) and only then closes them — same-target XOR writes commute, so open-then-close never leaves an instant with no mask on the wire. Without it the wire sat bare, holding its plaintext value, until its next filler (§5.6: every interior bare interval of the earlier builds came from this — ~ 450 per build, median $\sim 10^4$ gates). The same guard keeps every wire at **min_open** (2) masks, not just one: a burst opens as many replacements as needed before its closes, a filler or straddle open on a thin wire is followed by further opens, and a read on a thin operand keeps that many of its top-ups as real masks (§4). Every LGI sample (fillers, straddle opens, replacements, read top-ups) also draws its band wires **disjoint from every band wire already used by the wire's open masks**. Any shared wire biases the mask sum: two identical pairs cancel ($1 \oplus y \oplus xy$ twice is 0 — the wire is functionally bare while the bookkeeping counts two masks), two identical balancing wires cancel ($z \oplus z = 0$, no uniform term left), and a balancing wire equal to another open pair's wire folds the linear and the quadratic term into an OR ($x \oplus \neg x \wedge y = x \vee y$, biased 3:1). Each of these showed up as a $\phi \approx 0.25$ segment population in the gauntlet at 32 band wires (§5.7); at 256 the last one still hits $\sim 10\%$ of opens. The same discipline applies to the **burst gates**: a burst $b \hat{=} \text{lit}(c_1) \wedge \text{lit}(c_2)$ with a data control reads that wire under its masks, so its other control is drawn away from that wire's mask wires (and two data controls may not share a mask wire) — otherwise the product strips a mask's uniform term and the burst's flip correlates with the plaintext (ϕ 0.14–0.26, seen at **max_open** 2, §5.7).

Correctness is verified exhaustively over all 2^n inputs $\times$ many band settings for $n \leq 6$, $K \in 2..5$, all `max_open` and both rerand kinds (`scratchpad/v6`: 891 gadgets, 0 mismatches, plus a 360-case reorder-preserves- A check; the linear read), by the unit test in `blinded_v5.rs` ($n = 6$, all 2^n inputs $\times$ 8 band states $\times$ 6 seeds, **both read modes**), and end-to-end in `gen_sandwich_gadget` (forward + reverse-honesty sample-verify PASS in both modes).

3 Hidden firing

An *atomic*, mask-restoring gate module would leak which gate fired: the active wire’s before/after XOR across it equals the true increment $\Delta = comp \oplus lit(a) \wedge lit(b)$. That per-module increment is a robust invariant that survives the downstream mixing, so an adversary who segments the mixed circuit at module boundaries reads off A ’s gate list.

The fix uses the abundant masking gates on the *same* active wire: split the fire and emit one of c ’s LGI **opens between the halves**. It toggles c mid-fire and stays open past the module, so the net XOR on c is $\Delta \oplus (\text{secret band-mask})$ — never the bare increment. Because everything is co-sampled, the straddling LGI is a real, rerand-protected scaffold LGI opened exactly when the gate is placed, so **every** gate is covered, reusing the wire’s $u_w + 1$ budget at **no size cost** (measured: 7920/7920 gates on the $n=128$ sandwich). Earlier attempts — A -gate weaving ($\sim 23\%$), an injected firing mask (100% but +43% size), straddling fixed-scaffold slots (60–70%, tail slot-starved) — are superseded by generating the slot on demand.

4 Parameters and why

knob	prod.	meaning	why this value
K	2	band wires per LGI = per-LGI mask width ($\lfloor K/2 \rfloor$ pairs; $ \rho \approx \text{max_open} \cdot K$) — not the temporal length	affine/deg-2-neutral across K (§5); smallest is best. Size $\sim$ linear, read cost quadratic in $ \rho $, so large K explodes. Odd K wastes a wire.
max_open	3	rolling cap on simultaneously-open LGIs per wire	wider ρ = more local hiding, but read cost is quadratic in $ \rho $; 3 is the knee.
quad_fire	on (2026-09-06)	read operands from inside their quadratic masks; never linearise	the linearised read leaves the operand exactly affine in band wires for a window that every reordering stage stretches into the C -vs- G ridge (§5.0); quad-fire has no window, keeps the ridge at the I/O fringe through the whole pipeline, and is $\sim 12\%$ smaller. BV5_QUAD_FIRE=0 = legacy.
balanced	on (2026-09-07)	every LGI (and every read top-up) adds one CNOT $w \hat{=} z$ from a fresh band wire, so each mask term is $z \oplus 1 \oplus \neg x \wedge y$ — unbiased, still quadratic; band seed $x_i \oplus x_j$	a bare g57 term is 1 three times in four, so a wire under one open mask is <i>linearly correlated</i> with its plaintext (ϕ 0.29 with the C gate’s firing predicate; §5.6). Balanced masks take that channel to the null floor (median ϕ 0.08) at +92% gates ($K=2$, max_open 3: read polynomial $8 \times 8 \rightarrow 11 \times 11$ monomials); it is what passes the gauntlet’s w1/w2/w3 (§5.7). BV5_BALANCED=0 = plain masks; max_open 2 balanced is the +10% variant, weaker against two-feature scans.
min_open	2 (2026-09-07)	minimum open masks per data wire at <i>every</i> instant between its first and last mask; enforced wherever the count can drop or start low (burst replacements, filler/straddle opens, read top-ups kept)	one open mask is one uniform term, which one visible monomial cancels (§5.7); before the rule 26.5% of covered wire-time (3% on the payload half; 2,398 interior one-mask stretches, median 10.8k gates) sat at one mask. At 2: 0.0% and no interior stretch, for +0.03% gates (extra opens are offset by reads that need no top-up). Must be $< \text{max_open}$.
min_mask	auto = max_open = 3	hard floor per operand read: quadratic mask terms (quad-fire) / masking wires $ \rho $ (legacy)	guarantees no operand is ever read under fewer than 3 masks, even in a rare draw where the open pairs cancel (legacy: worst read $ \rho $ rises $2 \rightarrow 4$; mean $ \rho \approx 5.9$ unchanged; +0.2% gates).

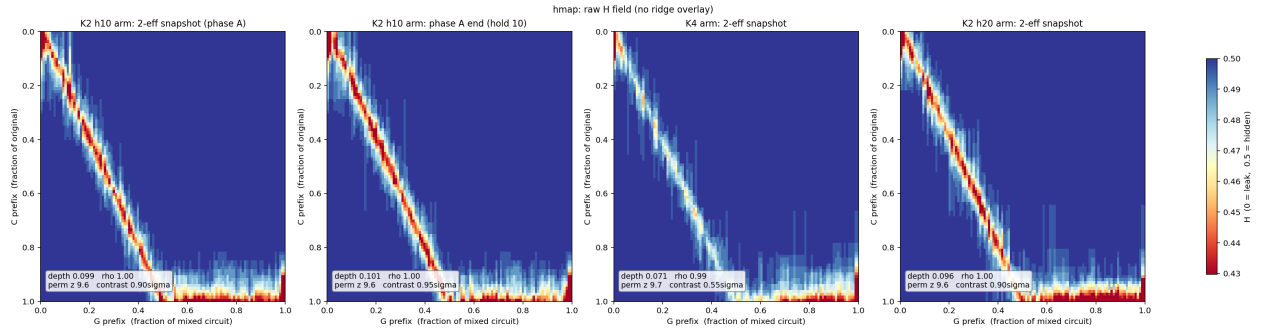
K sets each LGI’s mask *width*, not its temporal length — the identity’s circuit-spanning length is a property of the whole scaffold (§1e).

5 Measurements

All measurements are on the current co-sampled build with **burst rerand + the masking floor** ($n=128$, $|C| = |D| = 3000$, $K=2$). They match the pre-firing-fix baseline to within noise — neither the hidden-firing redesign nor the burst/floor refinements moved them (the burst change touches only the band wires, so it cannot move the data-wire ridge): degree-1 meanH = 0.4966, degree-2 meanH = 0.4967, exposed- $C < 0.35 \approx 5.6\text{--}5.7\%$.

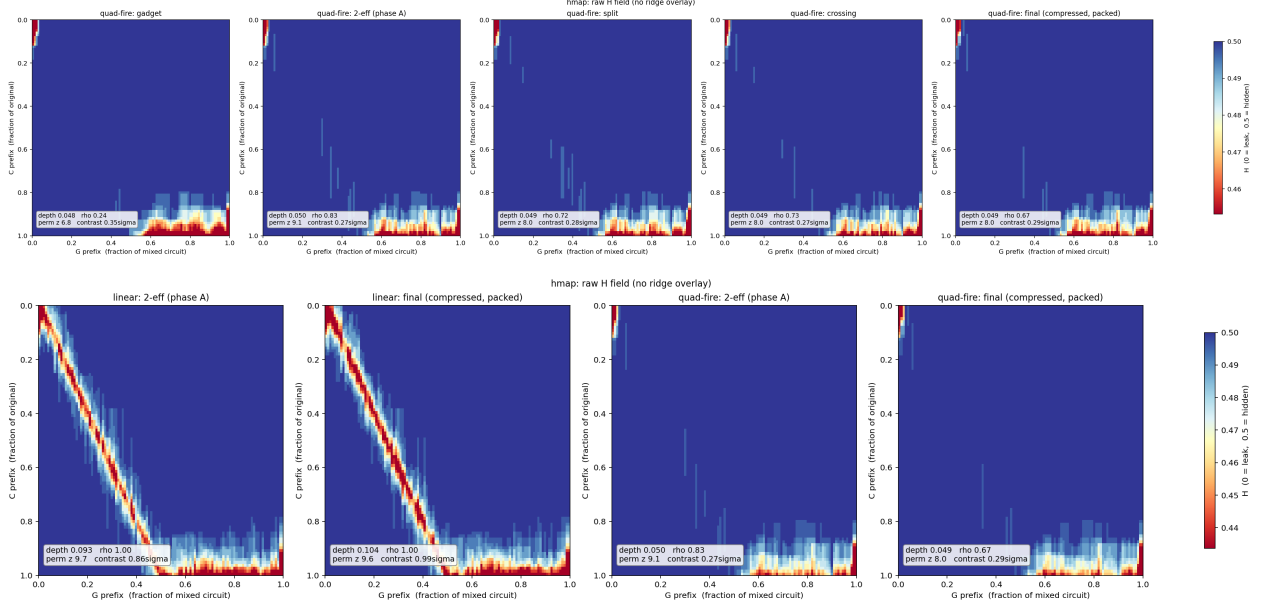
5.0 The affine ridge through the pipeline, and why the read mode changed (2026-09-06)

The figures in §5.1–§5.5 were read by **mean H** and by the count of **exposed rows** (segments with a cell below 0.25/0.35). Both are blind to a ridge made of *exact* affine relations at a few bits per cell, and the ridge statistic of `plot_hmap_ridge.py` (depth, median-row depth, ρ) had never been computed for the pipeline circuits. Computing it showed a full-strength diagonal in every mixed circuit of the linear-read design: the raw gadget has $\approx 500\text{--}800$ exact relations (median-row depth 0.008 — the “depth ≈ 0.05 ” of §5.1 is the I/O fringe plus a faint trace), the 2-eff snapshot $\approx 6,000$ (median-row depth 0.07, $\rho = 1.00$, permutation $z = 9.6$), and the compressed, packed final $\approx 7,000$ (depth 0.10). Each relation is one C bit equal to the XOR of the data wire and its ~ 6 linearised band wires: a **masked-read window**. The undo of a read is a write to the operand wire, pinned only by that wire’s next read, so every reordering stage — the DB splice moves of phase A, fmix’s final uniform float, the crossing walk — stretches the window to the operand’s idle interval (isolated: reordering-only moves $503 \rightarrow 601$ in snapshots, the final float alone $\rightarrow 4,555$; DB moves saturate by ≈ 1 move/gate). No parameter fixes it: extra filler LGIs or repair slots dilute it by $\approx \div 2$ at $+35\text{--}55\%$ gates, mask width is irrelevant, and `max_open = 2` is worse (full tables in `RIDGE_QUADFIRE_20260906.md`).



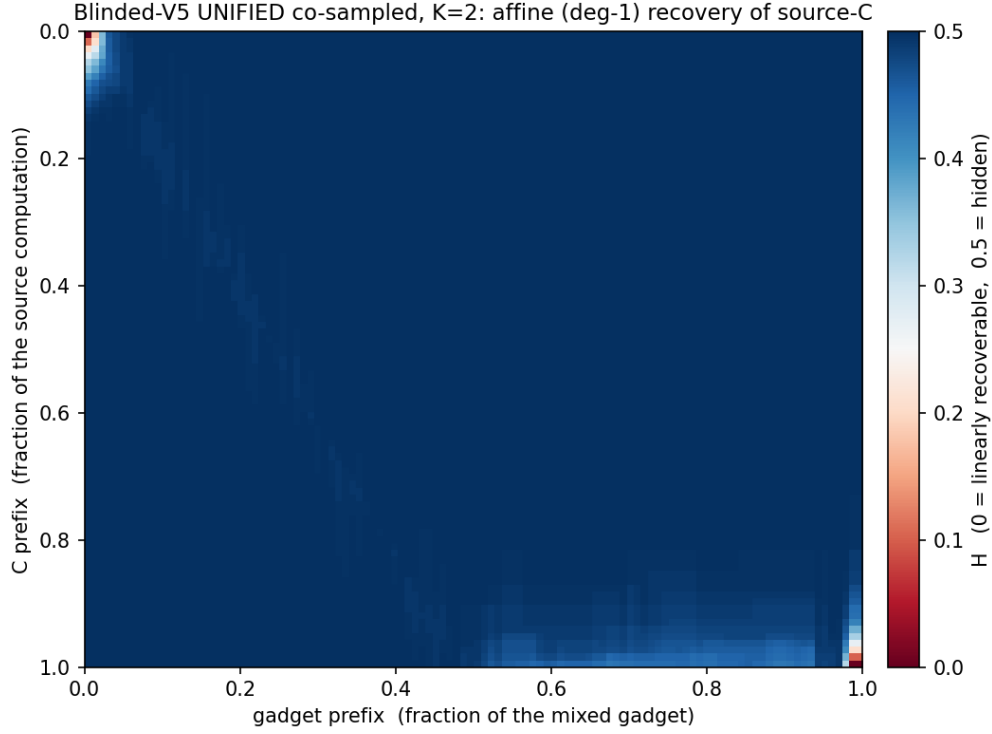
Quad-fire (§2 step 2) removes the window instead of diluting it. Same recipe, 2-eff pipelines, exact relations per stage (depth / median-row depth):

stage	linear read	quad-fire
gadget	563,356 g, 834 (0.049 / 0.008)	497,548 g, 500 (0.048 / 0.000)
2-eff phase A	962,429 g, 6,032 (0.093 / 0.066)	898,416 g, 443 (0.050 / 0.004)
split	1,839,248 g, 5,984 (0.096 / 0.070)	1,507,931 g, 478 (0.049 / 0.004)
crossing	3,345,382 g, 4,574 (0.081 / 0.051)	2,738,434 g, 368 (0.049 / 0.004)
final (fcompress + pack)	337,194 packed, 7,179 (0.104 / 0.082, ρ 1.00)	311,285 packed, 400 (0.049 / 0.004)



The residual 150–500 relations of quad-fire are single-wire identities in the delivered-output rows and at the raw input port — the public I/O fringe of §5.2 — and they do not grow with mixing (flat from the gadget to 3 moves/gate in phase A and through split, crossing and compression). The degree-2 adversary of §5.3 (products over a 24-data + 24-band slice) is likewise flat for quad-fire (141 $\rightarrow$ 102 after a float) where the linear read goes 481 $\rightarrow$ 4,978; a full degree-2 adversary with every band-pair product would cancel three open quadratic masks in either design (the existing sample-count-bounded posture), so quad-fire closes the affine channel, which was the open one.

5.1 Affine ridge — raw gadget, linear read

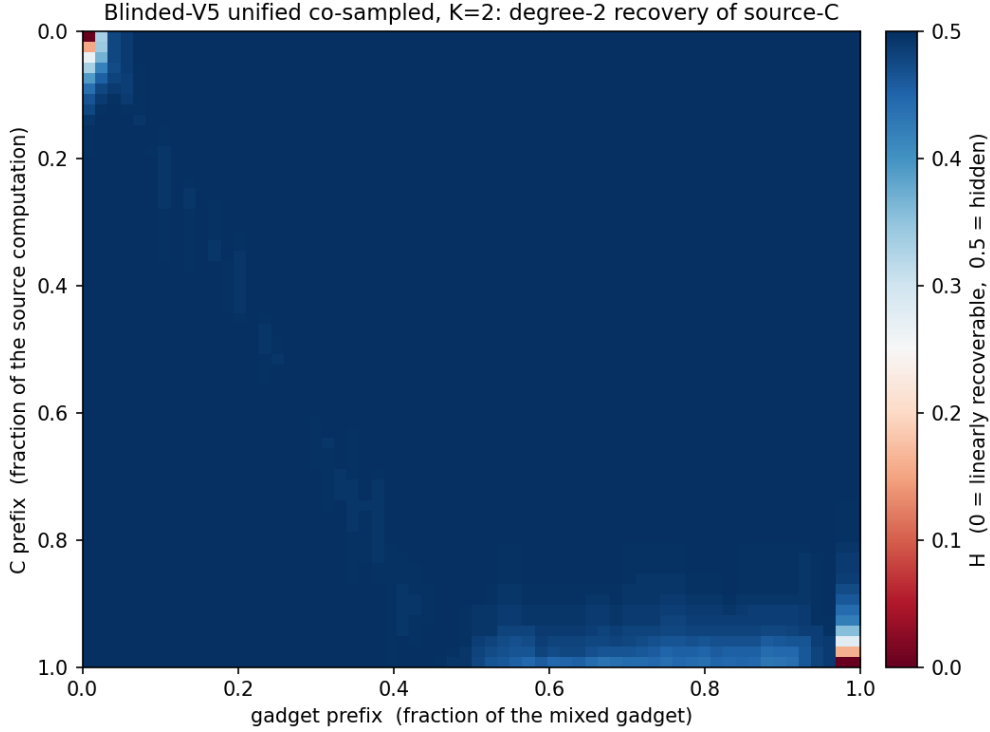


mean $H \approx 0.498$, ridge depth ≈ 0.05 ; the interior ($\sim 78\%$ of rows) is fully hidden ($H \approx 0.49$), and the only recovery is a thin fringe at the two endpoints.

5.2 The exposed fringe is public I/O — input segments are not leaked

The $\approx 6\text{--}10\%$ of source- C segments that are linearly exposed ($\text{min-}H < 0.25\text{--}0.35$) are the two corners of the diagonal. The **input side is masked as soon as the compute begins** — recovery is confined to the extreme top-left corner, i.e. the raw input port where x is public *before* any masking. The output side is the necessarily-delivered $C(x)$. So the input wire segments show a **lack of linear exposure** beyond the trivial public input; the boundary-cushion knobs measured no effect because the fringe is unavoidable public I/O, not a leak.

5.3 Degree-2 recovery



meanH ≈ 0.497 , depth ≈ 0.05 . The quadratic adversary reaches only ~ 0.01 deeper than affine, into the same low-degree I/O fringe; the interior stays fully hidden. Degree-2 is flat across K , like affine.

5.4 Rerand: bursts, straddle vs repair ($n_A = 256$, affine deg-1)

Each rerand slot is a **burst** of $F \approx 8K$ gates on one band wire; the **slot** count — not the gate count — drives the effect, since a straddle slot closes the masks reading b exactly once per slot. An early single-gate-dose study fixes where the thinning knee is (dose = total straddle/repair *gates*):

straddle	repair	gadget	meanH	affDepth
0	0	670k	0.4979	0.0550
1000	3000	668k	0.4959	0.0514
8000	0	607k	0.4878	0.0681
0	8000	780k	0.4985	0.0549

Heavy STRADDLE thins (meanH $\downarrow$, ridge $\uparrow$, gadget *shrinks* as masks close); heavy REPAIR does not (both = baseline, gadget *grows* from the pre/post pairs). The production plan keeps straddle **slots** on the safe side of the ≈ 1024 knee: auto $\approx m/(4K)$ (~ 875 for the half-size sandwich) $\times F = 8K \approx 2m$ band-refresh gates, **no repair**. Fewer, fatter slots mix the band as hard as many single updates while interfering less with the LGIs (and give the band its data-wire signature), and — because the thinning is driven by the *slot* count — they sit at the baseline floor (deg-1 meanH = 0.4966, deg-2 0.4967).

5.5 Through the GSS pipeline ($K = 2$ vs $K = 4$)

Four arms through `gss_mix.sh --gadgetization-mode blinded-v5` ($n=128$, half-size $|C| = |D| = 3000$, `--expand 2`, fresh independent CSPRNG seeds), measured at the **2-eff snapshot** (phase-A grow to $\approx 1.7\times$, an early mixing checkpoint; `hmap_affine --degree 1`, 92 C -segments):

arm	gadget g_{in}	2-eff gates	meanH	exp. < 0.25	< 0.35
$K=2$, hold 10	563,946	962,598	0.4945	6/92	11/92
$K=2$, hold 20	564,394	969,380	0.4942	6/92	11/92
$K=2$, hold 30	564,674	960,670	0.4951	6/92	9/92
$K=4$, hold 10	1,362,010	2,300,885	0.4965	6/92	10/92

- **meanH** ≈ 0.494 – 0.497 for both K — affine-neutral across K through the pipeline, matching the raw-gadget measurements (§5.1) and the pre-firing-fix baseline. (This is the corrected build with burst rerand + the masking floor + the even-filler / no-drain schedule; the pipeline statistics are unchanged, as the band-only refinements predict.)
- The exposed- C count (6/92 at < 0.25 , ~ 9 – $11/92$ at < 0.35) is the public **I/O boundary** of §5.2, confirmed by the per-segment profile: min-H ≈ 0 only at the input port (raw x) and output port ($C(x)$), climbing onto the hidden plateau within $\sim 5\%$ of C .
- The three $K=2$ arms agree because the 2-eff snapshot **precedes the hold phase**, so it is hold-independent — a seed-level consistency check.

Mean H and the exposed-row count do not see the ridge: these same 2-eff snapshots carry $\approx 6,000$ exact relations (depth 0.09–0.10, $\rho = 1.00$; §5.0). The arms in this table use the linear read; quad-fire arms are the follow-up. End-of-pipeline numbers for the linear read (post split / crossing / fcompress) are in §5.0’s table (2-eff pipelines) and in `RIDGE_QUADFIRE_20260906.md`.

5.6 Linear correlation with C ’s firing predicates; idle-bare intervals (2026-09-06)

The affine measures above are *exact* GF(2) tests; they are blind to a wire that is merely *statistically* close to a source value. `fire_corr` (`red_team_tests/bin/leakage/fire_corr.rs`) measures the phi (Pearson) correlation between each C gate’s **firing predicate** ($comp \oplus \bigwedge lit$, over 4096 random inputs) — or, with `--c-segments`, each C **state bit** — and every G segment and every G -gate increment, against a shuffled null (null max ≈ 0.10). Calibration: a fire vs its *bare* operand is phi 0.577; vs the operand under one `g57` mask term 0.289 (the term $1 \oplus \neg x \wedge y$ is 1 with probability $3/4$); two terms 0.144. Read on the raw gadget ($n = 128$, same C and seeds throughout; “interior” = C gates in the middle 70% of C , which excludes the public I/O fringe; counts out of 3000 C gates, 2101 interior):

build		gates	interior idle- bare intervals	fires: int. median phi	≥ 0.3	≥ 0.5	state bits: int. ≥ 0.5	exact copies
quad-fire, the coverage (0fb96997)	before the fixes	496,386	448 (med. 11k)	0.289	907	19	336	1
balanced, fixes	before the fixes	939,214	735 (med. 19k)	0.079	98	67	38	13
quad-fire fixes	+ coverage	514,502	0	0.287	907	15	315	0
balanced fixes (default)	+ coverage	984,384	0	0.078	31	15	9	0
null (shuffled)				0.077	0	0	0	0

Three findings.

1. **The one-mask population.** Half of C 's gates have a segment on an operand wire at $\phi \approx 0.29$ = the operand under a *single* biased mask term: masking depth between reads is typically 1–2 terms, and each term is biased. This is the channel **balanced** closes — the interior median drops from 0.29 to the null floor (0.078) and the state-bit hits from 315 to 9 — at +92% gates. It survives the pipeline unchanged (quad-fire 2-eff: 862 interior ≥ 0.3 ; final 642), because reordering does not change a wire's statistics.
2. **Idle-bare intervals (a build defect, fixed).** The strongest interior hits ($\phi \approx 0.58$ = *bare* operand) were exact copies of C state bits sitting on a wire for thousands of gates. A rerand burst on band wire b closes every open LGI reading b ; when that was the wire's *only* open LGI the wire was left bare until its next filler open — every interior bare interval of the earlier builds (448 / 735, all of them) had this trigger, and balanced LGIs (three band wires instead of two) were hit 50% more often. The fix is the *cover-replacement* rule (§2: open a fresh LGI on the wire before the close; ≈ 430 (plain) / ≈ 780 (balanced) replacements per build, $\approx 0.3\%$ gates). A second, smaller population were seldom-used (high-half) wires whose only masks were read-time top-ups, undone after each fire; the *read-cover* rule keeps the first top-up open (~ 115 – 125 per build, no cost). A third, ~ 1 per build, was two identical open pairs cancelling; LGI sampling now avoids a wire's open pairs. After the three rules the census finds **no** interior bare interval in either variant, and the remaining ≥ 0.5 hits (15, identical in both variants) sit at the ends of the interior window next to the public I/O fringe. The pre-fix defect is in every earlier blinded-V5 deliverable (incl. the $K=2$ 2-eff files); reordering stretches those bare windows (interior ≥ 0.5 : gadget 19 $\rightarrow$ 2-eff 81 $\rightarrow$ final 52).
3. **What the scans do not see.** The band itself is nearly uniform after the bursts ($|P - \frac{1}{2}| \approx 0.02$), the AND seed ($x_i \wedge \neg x_j$, bias $\frac{1}{4}$) versus the balanced XOR seed makes no difference to the residual, and restricting bursts to band-only controls (BV5_BURST_BANDONLY=1) does not help either — the defect was structural, not statistical.

Diagnostics: `fire_corr --probe <C gate list>` prints, for each listed C gate, its best fire-partner segment's ϕ against the gate's $(a, b, c_{\text{old}}, \text{fire}, c_{\text{new}})$ (the gauntlet's 5-tuple), which is what identified the bare copies (a : +1.000); BV5_DIAG=1 reports the replacement and read-cover counts.

5.7 The gadget gauntlet (colleague's trace-and-audit battery), $n = 32/64$ (2026-09-07)

`tests/gauntlet` records, for random inputs, every wire's initial value and every gate's flip and new target value of a gadgetized r57 chain, and runs six attacks against the chain's $(a, b, c_{\text{old}}, \text{fire}, c_{\text{new}})$

per source gate: **a1** (a feature equals a target), **xrows** (a target is affine in one prefix’s wire state), **xtrace** (affine in the whole trace), and **w1/w2/w3** (correlation with one feature, or an XOR/AND/OR/AND-NOT of two or three, capped strided subsets), flagged above the NULL column’s maximum and 6σ . Blinded-V5 enters as native arms (**blindedv5** = plain masks, **blindedv5_balanced**): production preset, **encoded I/O** (**encoded_io**: **max_open** masks per wire pre-opened and the drain skipped, the encode/decode gates applied out of band — the convention of the colleague’s share-encoded arms, so no raw input or output is a wire value), band $\max(32, n)$ wires starting uniformly random. The harness’s own controls (**none**, **secretshare14**, **bandproduct92**) light up every attack. Numbers are nontrivial hits / flags out of $5k$ targets (k source gates):

n	k	arm	mix	a1	xrows	xtrace	w1	w2	w3
32	64	balanced	off	0	0	320	0	0	0
32	256	balanced	off	0	0	1280	0	0	0
64	64	balanced	off	0	0	320	0	0	0
64	256	balanced	off	0	0	1280	0	0	0
32	64	quad-fire (plain)	off	0	0	320	320	27	35
64	256	quad-fire (plain)	off	0	0	1280	1280	5	5
64	256	none control	off	1096	840	1096	1280	1214	442
32	64	balanced	on	0	1	320	28	0	0
32	256	balanced	on	0	0	skip	47	0	0
64	64	balanced	on	0	0	320	0	0	0
64	256	balanced	on	0	0	skip	60	0	0
64	256	quad-fire (plain)	on	7	7	1280	1255	23	13

Reading it:

- **Unmixed, the balanced build passes every discriminating attack** at both sizes and both chain lengths; the plain build fails **w1** on every target (the one-mask bias of §5.6) and **w2/w3** on a few. **xtrace** is an identity for XOR masking — every mask term is the flip of the gate that applied it, so each plaintext is an exact XOR of trace features — and fires for every XOR-based arm including the harness’s controls; only the colleague’s share-encoded **nonlinear193** passes it (nonlinear decode). It is not a statement about what an adversary without the targets can do.
- **The harness found three more generator gaps**, each a sampling coincidence that made a mask sum biased: two open masks on a wire sharing a pair, a balancing wire, or a balancing wire equal to another mask’s pair wire ($x \oplus \neg x \wedge y = x \vee y$); and, at a fire, a band wire shared between the target’s masks and the operands’ polynomials (phi 0.1–0.25 segments mid-fire). The disjoint-sampling rule (§2) and the fire-cover bracket (§2, step 4) closed them; the bracket costs $\approx 3\%$ gates.
- **Mixed cells are the harness’s own mixer** (store-free crossings, copy splits and conjugation twists at 20k moves), not the pipeline: its splits break the quadratic masks into CNOT-sized pieces and reorder them, so intermediate states carry partial masks. That produces sporadic **w1** flags on the balanced arm (max phi 0.4–0.5; none at $n = 64, k = 64$) and, on the plain arm, exact plaintext copies (**a1** 7 at $n = 64, k = 256$). The pipeline’s own reorderings are measured end to end with **fire_corr** instead (§5.6).
- At $n = 8$ (the colleague’s default) the picture is the same, with the encoded-I/O convention the only way to make the comparison meaningful: with plaintext I/O nearly every target is an input or output at that size.

$n = 256$ (**V5 input width 256, band 256**), **balanced, unmixed**, `max_open 3` versus the size-neutral `max_open 2` (`blindedv5_balanced_mo2`):

k	<code>max_open</code>	<code>a1</code>	<code>xrows</code>	<code>w1</code>	<code>w2</code>	<code>w3</code>	note
256	3	0	0	0	0	0	
512	3	0	0	0	0	0	
256	2	0	0	$4 \rightarrow 0$	6	0	<code>w1</code> before $\rightarrow$ after the burst-control rule
512	2	0	0	0	$20 \rightarrow 0$	0	<code>w2</code> before $\rightarrow$ after the burst-control rule

Two more things came out of it. (i) A **burst gate** whose controls are a masked data wire and that wire’s own balancing wire strips the mask’s uniform term $((x \oplus z \oplus q) \wedge \neg z)$: the burst’s flip correlated with the plaintext at ϕ 0.14–0.26 — the sixth and last sampling rule (§2: burst controls avoid a data control’s mask wires). (ii) **`max_open 2` is not equivalent to 3**. With a single open balanced mask $x \oplus z \oplus q$, the XOR of the wire’s segment with any visible monomial that contains z (a fire monomial or a burst reading z) cancels the uniform term and leaves the biased quadratic part (ϕ 0.23); the harness’s capped two-feature scan finds it when its strided subset happens to hold such a pair (`w2` 6 of 1280 at $k = 256$; 0 at $k = 512$ after the burst rule) and its mixer wrecks it (`a1` 10, `xrows` 80, `w1` 201 at $k = 256$ mixed, versus `w1` 6 / `w2` 1 for `max_open 3`). Three open masks carry more independent uniform terms than a pair of features can cancel. So the +10% variant buys the `w1` result but not the multi-feature one; `max_open 3` at +92% is the clean build.

6 Tradeoffs and the current choices

- **Mask width K vs size.** Larger K widens each mask, not the temporal length; hiding is flat in K while size grows $\sim$ linearly and the read cost quadratically, so **$K=2$** ; $K=4$ only for a wider read-mask.
- **Band turnover vs size (bursts, slots vs F).** More band updates = more entropy and stronger resistance to commutation-back, but **STRADDLE slots** thin the masking past ≈ 1024 and **REPAIR slots** grow the gadget. Bursting decouples the two knobs: F (gates/slot) sets how hard each band wire is stirred and how data-like its activity looks, while the **slot count** ($\approx m/4K$, held under the knee) sets the thinning. Auto $m/4K$ slots $\times F = 8K$, no repair, sits at the baseline floor.
- **Masking floor is a cheap hedge.** `min_mask` guarantees every operand is read under ≥ 3 masking wires (worst-case $|\rho|$ $2 \rightarrow 4$; mean unchanged); it fires only on the rare under-floor draw, costing $\sim 0.2\%$ gates — a hedge against a low-probability thin-masking event, not a routine cost.
- **Firing-hiding is free** in the co-sampled build (it reuses the existing LGI budget); earlier fixed-scaffold approaches paid coverage (23–70%) or +43% size.
- **The residual fringe is I/O, not a knob** — no parameter removes the public input/output boundary.
- **Coverage is enforced, not budgeted.** The u_w+1 budget sets the mask *statistics*; the six coverage rules (§2, §5.6: cover-replacement across a burst, read-cover on an uncovered operand, disjoint mask wires incl. duplicate-free pairs, the fire-cover bracket, the burst-control rule) make “no wire is ever bare between its first and last LGI” a property of the build rather than a likely outcome, for $\approx 3.5\%$ gates (the bracket is $\approx 3\%$ of it); `min_open 2` lifts the floor

from one mask to two for +0.03%.

- **Balanced masks (default since 2026-09-07).** The biased `g57` term leaves a wire under one mask linearly correlated with its plaintext (ϕ 0.29); **balanced** removes it (null floor) for +92% gates at $K=2/\text{max_open}$ 3 and is the build that passes the gauntlet’s correlation battery (§5.7). The exact-affine measures are unaffected either way, and the cost is the read polynomial, $(3\text{max_open} + 2)^2 = 121$ monomials per fire instead of $(2\text{max_open} + 2)^2 = 64$. `max\_open= 2` **balanced** brings the polynomial back to $8 \times 8 = 64$: measured 565,374 gates at $n = 128$ (+10% over the plain `max\_open` 3 build’s 514,502, versus +92% for balanced `max\_open` 3’s 984,384), with the same `fire_corr` statistics as balanced `max\_open` 3 (interior median 0.076 vs 0.078, p90 0.137 vs 0.133, no bare intervals) — but the gauntlet’s two-feature scan separates them: one open mask is one uniform term, which a single visible monomial cancels (§5.7). `max\_open` 3 remains the clean build.
- **Read mode: quad-fire, not linearisation.** Linearising a read is the cheapest way to fire (degree-2 monomials only) but leaves the operand affine in band wires for a window that reordering stretches into the pipeline ridge (§5.0). Firing from inside the quadratic masks costs degree ≤ 4 monomials, needs no linearise/undo bracket, makes the gadget $\sim 12\%$ smaller, and keeps the ridge at the fringe end to end. Repair slots are counter-productive with quad-fire (they close masks around bursts); keep `max\_open= 3`.